Hall Ticket No: | | | | | | | | | | |

Course Code: A8520

Ashwini
CSE

VARDHAMAN
COLLEGE OF ENGINEERING

III B. Tech I Semester Continuous Assessment - II, October -2025

(Regulations: VCE-R22)

SOFTWARE ENGINEERING

(Common to CSE, CSE(AIML), IT)

Date: 22 October 2025 [AN] **Time: 2 Hours** **Maximum Marks: 30**

Answer all Questions in Part-A

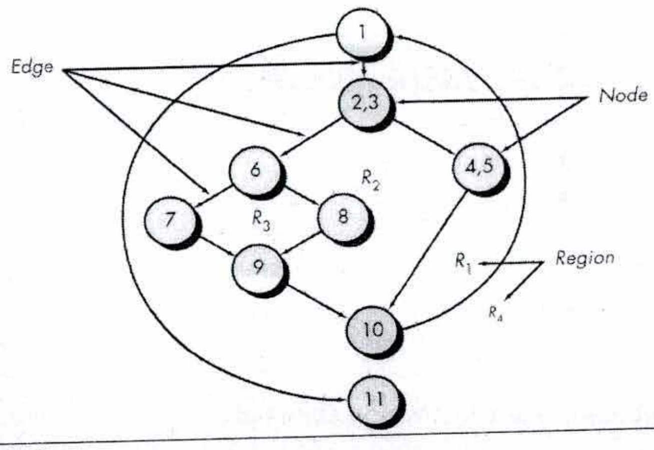
Answer any FOUR Questions in Part-B

Course Outcomes with Bloom's Levels:

CO#	***CO Statement***	***Bloom's Level (L#)***
CO1	Identify the design issues and process models to develop software.	L2
CO2	Determine the functional and non functional requirements with appropriate validation for a software product.	L5
CO3	Develop software design documents for the given requirements.	L6
CO4	Prepare test documents at various stages to validate the project.	L6
CO5	Illustrate the need of quality management and metrics for product standardization.	L2

PART-A (Multiple Choice / Fill in the Blanks / Match the Following / Short Answer Type Questions)

			CO#	***BL#***	***Marks***
1.	a)	Write a short note on Interaction Diagrams	CO3	BL2	***1 M***
	b)	OOD Stands for ______________	CO3	BL1	***1 M***
	c)	What are structural Diagrams?	CO3	BL1	***1 M***
	d)	Compare system testing and integration testing.	CO4	BL4	***1 M***
	e)	Discuss an example of top-down integration	CO4	BL3	***1 M***
	f)	What is the primary purpose of debugging?	CO4	BL3	***1 M***
	g)	What is the first step in a strategic approach to testing	CO4	BL1	***1 M***
	h)	What is formal technical reviews.	CO5	BL2	***1 M***
	i)	Why are failure costs usually higher than appraisal costs?	CO5	BL4	***1 M***
	j)	How does ISO 9000 certification influence customer satisfaction in a software company?	CO5	BL2	***1 M***

PART-B (Descriptive Questions)				
2.	Explain the importance of making the user interface consistent. Discuss how consistency in interface design improves usability and user experience, with suitable examples	CO3	BL4	**5 M**
3.	Write a note on Use Case Diagram and draw the Use Case Diagram for the Library Management System.	CO3	BL3	**5 M**
4.	Write independent paths for the following graph 	CO4	BL4	**5 M**
5.	Explain stub and driver programming with a suitable example.	CO4	BL3	**5 M**
6.	Explain four activities for achieving software quality. Illustrate your answer with examples	CO5	BL4	**5 M**
7.	Break down the main elements of the ISO 9001 quality management system and analyze how each element ensures product and process quality in software development..	CO5	BL4	**5 M**

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SCHEMA OF VALUATION

Name of the Subject	: Software Engineering	Subject Code	: A8520
Programme	: B.Tech.	Year	: III yr I sem
Examination	: Regular / Supplementary	Date of Examination	: 22/10/25
Max. Marks	: 30	Time of Examination	: FN.

Part-A

1.(a) Interaction Diagrams: used to model the dynamic behavior of a system, they show how objects interact with each other through messages to carry out a specific task } 1M

(b). Object Oriented Design. } 1M

(C) Structural Diagrams:
Represents how the system is organized rather than how it behaves. } 1/2 M

Examples: 1. class Diagram
2. Object "
3. Component "
4. Deployment "
} any two 1/2 M

(d)

System testing	Integration Testing
1. Focuses on overall system behavior and functionality	1. Focuses on data flow and interface between modules
2. usually done by independent testers	2. done by developers

1/2 M 1/2 M

(e) Top down Integration Testing starts from the top-level module and moves to lower-level modules step by step. } 1M

Ex: Banking system

(f) Debugging:

primary purpose of debugging is to identify, analyze and remove error in program so that it functions correctly according to its specifications. } 1M

(g) The first step in the strategic approach to software testing is a planning
→ It involves deciding what to test, how to test, when to test and who will test. } 1M

(h) Formal technical reviews:

is a structured and planned meeting in which a team of software engineers examines the design, code or documentation of s/w product. } 1M

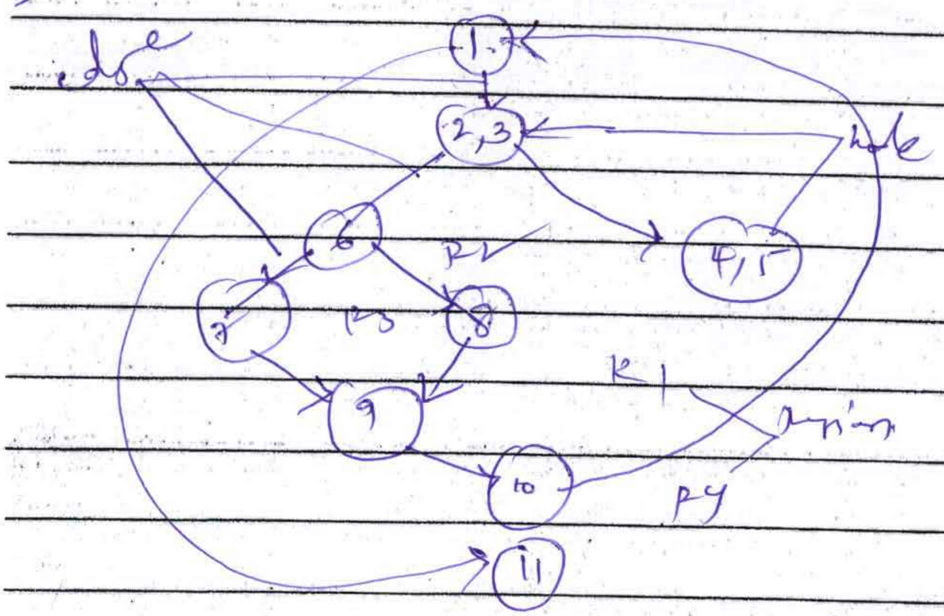
c(i) why are failure costs higher than appraisal cost

Appraisal cost: cost of finding defects early.
Failure cost: cost of fixing defects late.
Ex: finding and fixing a bug during coding is cheap, but fixing it after release may require patch updates customer support which is expensive.

(f) It ensures that a software company follows internationally recognized quality Management Standards', it focuses on consistent processes, continuous improvement and all of which directly enhance customer satisfn

4. Ans

Given graph.



1M

An Independent path is any path through the program that new set of processing statements or a new condition

Independent program paths

2M

path 1 : 1-11

path 2 : 1-2-3-4-5-10-1-11

path 3 : 1-2-3-6-8-9-10-1-11

path 4 : 1-2-3-6-7-9-10-1-11

3M

(2) A Consistent user interface means that similar elements behave, look & feel the Same way throughout an application or website. Consistency ensures that users can easily Predict what will happen they Perform an action, leading to better usability, learnability & user satisfaction.

Why Consistency is important:-

1. Improve usability
2. Enhance user Experience
3. Reduces Errors
4. speeds up task performance
5. Supports Branding & Aesthetic Harmony

3m

Types of Consistency:-

Type	Description	Example
Visual Consistency	Same colour, fonts, icons, Layouts	All "submit" buttons look like
Functional Consistency	Same actions Produce Same results.	Clicking "search" always displays Search results.
Internal Consistency	uniformity across diff with a single Product	Menus & Shortcuts behave the Same on all Screens
External Consistency	uniformity across different Products.	copy-paste shortcut work in all apps.

2m

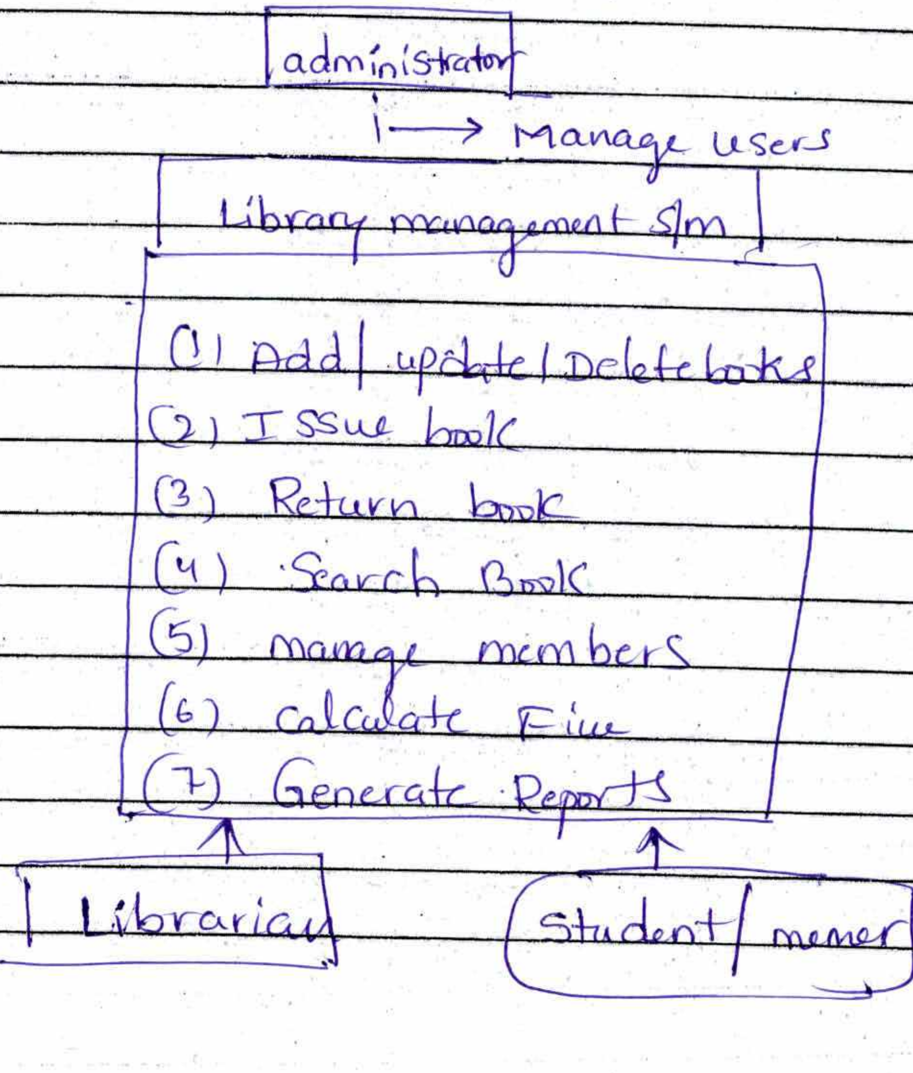
(3) Note on user case Diagram:- A use case diagram is apart of the unified Modeling Language (UML) used to represent the functional requirements of a s/m. 2M

It shows how different users (actors) interact with the system and the various functions (use cases) it performs.

Purpose of use case Diagram:-

* To Identify the main functionalities of a s/m
* To show the interaction b/w user & system.
* To serve as a communication tool b/w stakeholders & Developers.

User case Diagram for Library management s/m:-



3M

5

In software testing, especially during integration testing, all modules of a System may not be ready at the Same time

1. Stub:-
Definition:- A stub is a dummy module that Simulates the behaviour of a called module.
It is used during top-down integration testing.

2. Driver:- A driver is a dummy module that Simulates a calling module.
It is used during bottom-up integration testing. 3m

Comparison:- 2m

Aspect	Stub	Driver
used in	Top down Integration	Bottom-up Integration
Replaces	called Module	calling module
Purpose	Simulate lower-level module	Simulate higher-level module
Example	Dummy function returning Sample output	Temporary main() calling the function.

(6) ~~Software Quality means ensuring that a s/w Product meets the Customer's requirements, is reliable, maintainable, and error free~~

~~To achieve high quality, several activities are carried out throughout~~

(6) Achieving high software quality involves a combination of planning, verification, validation and continuous improvement — 1m

1. Requirement Analysis and specification
2. Software Design and code Reviews
3. Testing (Validation)
4. Configuration Management and continuous improvement

4m Expl

(7) 1. Context of organization
2. Leadership & commitment
3. planning
4. Support
5. operation
6. performance Evaluation
7. Improvement

explanation 5m

VCE